Geometry-MMALS G1 v1.0.3

Cross-Angle Paired Geometry Protocol Correction Report

Status: protocol release and GitHub archive. Accepted claim: C0 implementation only. C1-C6 remain unqualified.

This report records the scientific diagnosis of the v1.0.2 development run, the protocol correction implemented in v1.0.3, the mathematical and engineering changes, and the conditions required before any Geometry-MMALS claim.

Release	v1.0.3
Date	2026-06-13
Primary notebook	Geometry_MMALS_G1_CrossAngle_Paired_v1_0_3.ipynb
Repository	gharbonnier78/geometry-mmalls-g1
Epistemic status	Development protocol; not a qualified result

1. Executive conclusion

v1.0.2 solved a real numerical problem: the route geometry loss no longer corrupted the model with NaN gradients. The development run completed with finite traces, finite ablations and a finite tangent probe. Nevertheless, it did not constitute a valid test of grounded angular geometry.

The reason is structural rather than statistical. Each training batch came from one fixed-angle loader. Every factor value inside the batch was identical, so the loss could enforce consistency within an angle but could not compare distinct angles or learn their order.

v1.0.3 changes the unit of optimization and analysis to the same MNIST source image observed under several angles. The corrected notebook therefore makes the core G1-A question measurable: do source-matched changes in angle induce coherent changes in context, route, host function and synthesis?

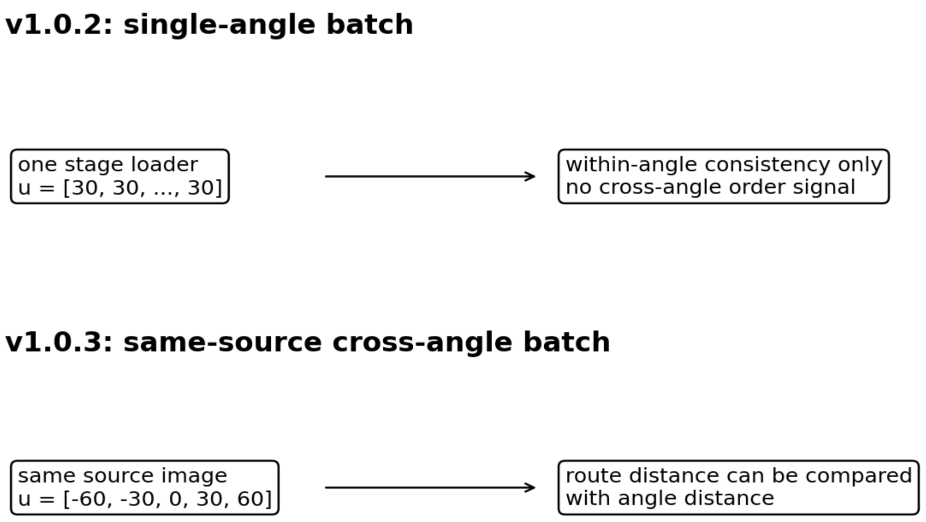

Figure 1. The protocol correction. Interpolation and extrapolation angles remain excluded from training.

2. Research history

Version	Primary purpose	Outcome
v1.0.0	Specify functional geometry and a falsification ladder.	Article and C0 scaffold.
v1.0.1	Audit claims, gates, notation and memory interfaces.	C0-only status formalized.
v1.0.2	Repair route-loss NaNs and Colab execution.	Finite development run.
v1.0.3	Correct the cross-angle optimization and measurement unit.	Paired protocol implemented.

3. What the v1.0.2 development run showed

The v1.0.2 run is retained as a negative and diagnostic development artifact. Its finite measurements are useful, but none supports the C1 grounded-geometry gate.

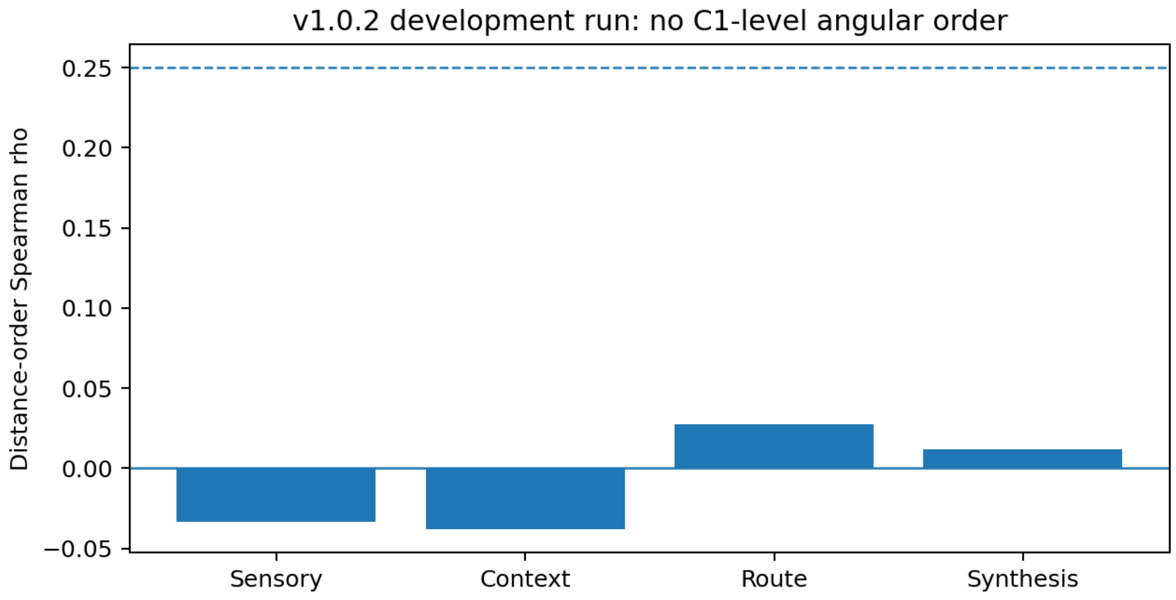

Figure 2. Distance-order correlations from v1.0.2. The dashed line marks the pilot rho threshold of 0.25; confidence and control conditions were not yet valid.

Space	rho	Stress	Scientific reading
Sensory	-0.0334	0.6070	No measured angular order.
Context	-0.0383	0.7598	No measured angular order.
Route	0.0272	0.7084	No measured angular order.
Synthesis	0.0118	0.6054	No improvement beyond sensory.

4. Why those measurements were non-diagnostic

Single-angle loss. Inside each training batch, all angle differences were zero. Far-angle separation was never activated.

Identity-angle confounding. Global distances mixed changes in digit identity with changes in rotation. The correct primary comparison is within the same source image.

Tied neighborhoods. Thousands of samples shared exactly the same angle. Ordinary sample-level kNN selected arbitrary members of large tied sets and was not a valid primary geometry gate.

Dependent p-values. Millions of pairwise distances were treated as independent. Source-block bootstrap is required instead.

Unsigned causal norm. The v1.0.2 tangent probe measured only the magnitude of route change, so symmetric effects for positive and negative interventions were expected.

5. v1.0.3 protocol

For each source image x_i and the angles already seen at continual stage t, the dataset returns a tensor of source-matched views. The router never receives the angle as input. The angle is available only to the declared supervised geometry loss and evaluator.

Classification: current-angle view by default. Geometry: all seen-angle views of the same source. Held-out angles: evaluation only.

5.1 Optimization geometry

The paired loss uses chordal distance in square-root simplex coordinates during optimization. This distance is monotonic with the Fisher-Rao geodesic and avoids the arccos derivative singularity. Fisher-Rao remains the evaluation metric.

L_pair = L_local + lambda_far * L_far + lambda_match * L_match

5.2 Measurement geometry

Distance-order correlation and stress are calculated independently inside every source-image block. Group-level values are then summarized with source-block bootstrap confidence intervals. Factor-centroid geometry provides a complementary tie-free view.

6. Tracked changes

ID	Change	Purpose
CHG-103-01	MultiAngleMNIST	Same source under several angles.
CHG-103-02	Paired route geometry loss	Real cross-angle training signal.
CHG-103-03	Seen-angle anchors only	Preserve continual chronology.
CHG-103-04	Source index in traces	Paired analysis and block bootstrap.
CHG-103-05	Source/centroid metrics	Remove identity confounding and tied kNN.
CHG-103-06	Interpolation controls	Test unseen intermediate angles.
CHG-103-07	Staged evaluation	Measure forgetting across angles.
CHG-103-08	Signed causal probe	Test angle-specific direction versus controls.
CHG-103-09	Version and git SHA	Prevent silent source mismatch.
CHG-103-10	Explicit non-claims	Preserve epistemic discipline.

7. Notebook outputs

The notebook exports staged training logs, per-angle accuracy and NLL, forgetting, paired-source geometry, factor-centroid geometry, held-out interpolation errors, host ablation summaries, signed causal controls and an immutable run manifest.

Artifact	Content
stage_logs.csv	Loss components by method and continual stage.
staged_evaluation.csv	Accuracy and NLL after every stage at every angle.
forgetting.csv	Best-to-final accuracy loss on trained angles.
paired_geometry_metrics.csv	Source-block bootstrap and centroid geometry.
interpolation_metrics.csv	Held-out performance and route/context interpolation.
host_ablation_summary.csv	Mean, median and positive contribution fraction.
causal_probe.csv	Signed tangent effect, orthogonal controls and identity drop.
run_manifest.json	Version, git SHA, split checksum, methods and non-claim status.

8. Gate status and non-claims

Gate	v1.0.3 status	Requirement before claim
C0	Implemented	Run archive, tests, environment and source verification.
C1	Measurable, unqualified	Multi-seed paired-source CI and controls.
C2	Measurable, unqualified	Beat nearest-route and no-geometry controls.
C3	Partly measurable	Localized contribution and cross-seed role matching.
C4	Development probe	CSR CI, signed monotonicity and identity guard.
C5	Not implemented	Explicit transport and dual-memory experiment.

Gate	v1.0.3 status	Requirement before claim
C6	Not evaluated	Tuned baselines and secondary-dataset replication.

No quantum, consciousness, universal-intelligence or positive backward transfer claim is introduced by this release.

9. Reproducibility and GitHub archive

The notebook checks package version 1.0.3 and records the active git commit SHA. This prevents a notebook-local function from silently masking an older installed package. The report, Markdown source and notebook are placed under versioned repository paths.

Repository path	Role
notebooks/Geometry_MMALS_G1_CrossAngle_Paired_v1_0_3.ipynb	Primary experiment notebook
docs/reports/Geometry_MMALS_G1_v1_0_3_Protocol_Correction_Report.pdf	Immutable archive report
docs/reports/Geometry_MMALS_G1_v1_0_3_Protocol_Correction_Report.md	Editable report source
src/geometry_mmalls/data.py	Same-source multi-angle dataset
src/geometry_mmalls/geometry.py	Paired route geometry loss
src/geometry_mmalls/metrics.py	Source-block and centroid metrics
tests/	Regression and measurement tests

10. Recommended execution sequence

1. Push or install the complete v1.0.3 package; do not run the notebook against v1.0.2.
2. Run the development mode with one seed and both no-geometry and paired-geometry variants.
3. Inspect finite-value guards, staged accuracy, paired-source geometry and held-out interpolation.
4. Freeze any revised pilot thresholds before qualification.
5. Run at least five seeds and archive every failed seed.
6. Harden EWC, replay, sparse-MoE, oracle-angle and joint controls with equal validation budgets.
7. Only then assess C1-C4. Do not infer C5 from the paired anchor mechanism.

11. Final research interpretation

v1.0.3 is not a positive-result patch. It is a protocol-validity patch. Its value is that a failed run will now be informative: the model will have received real cross-angle evidence, the metrics will respect source identity, and uncertainty will be summarized at an appropriate block level. A future positive result will therefore be more credible, and a future negative result will be harder to dismiss as a measurement error.